Modeling Allele Frequency Distributions in Human Populations

A Statistical Analysis of *HLA-B*, *CYP2C19*, and *APOE* using gnomAD v4 Data

GitHub(Modeling-Allele-Frequency-Distributions-in-Human-Populations)

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Contents

1 Introduction

Quantifying how allele frequencies fluctuate within and between human populations provides key insights into evolutionary mechanisms, mutation rates, and founder events. Theoretically, under a neutral Wright-Fisher model at equilibrium, the stationary distribution of allele frequencies under the joint effects of recurrent mutation and genetic drift follows a continuous Beta distribution:

$$f(x;\alpha,\beta) = \frac{1}{\mathrm{B}(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad x \in (0,1) \tag{1}$$

where the shape parameters are directly tied to the effective population size (N_e) and mutation rates (μ).

This report presents a full end-to-end data processing and statistical inference pipeline implemented in R to fit and critically evaluate the suitability of the Beta distribution model against empirical high-throughput sequencing records.

2 Data collection

Allele frequency data for the three genes of interest — *HLA-B* (Major Histocompatibility Complex class I), *CYP2C19* (Cytochrome P450 2C19), and *APOE* (Apolipoprotein E) — were obtained from gnomAD (Genome Aggregation Database, v4), a large-scale public repository of human genetic variation aggregated from sequencing studies across diverse populations.

The datasets were downloaded manually from the gnomAD browser:
(`https://gnomad.broadinstitute.org/`) by searching each gene by name and exporting the variant table as a CSV file. Each file contained one row per variant and included, for each of the ten reported ancestry groups, the allele count (AC), allele number (AN), and homozygote count. The ten populations provided by gnomAD are listed in Table 1.

Table 1: Ancestry groups provided by gnomAD v4.

Population Code	Ancestry Group
AFR	African/African American
AMR	Admixed American
ASJ	Ashkenazi Jewish
EAS	East Asian
FIN	European (Finnish)
MID	Middle Eastern
NFE	European (non-Finnish)
AMI	Amish
SAS	South Asian
REM	Remaining

Before conducting formal statistical modeling, a robust data preprocessing and reshaping pipeline was implemented in `R` utilizing the `tidyverse` ecosystem.

The primary structural challenge of the raw datasets was their horizontal, *wide* format. For each unique genetic variant, population-specific metrics (such as allele counts, allele numbers, and homozygote counts) were distributed across separate columns for all the populations.

To make this data computationally viable for downstream data visualization and statistical analysis, an automated pipeline was engineered to transform the data from a wide format into a standardized *long* format.

2.1 Pipeline Architecture and Data Transformation

The pipeline is anchored by a functional architecture designed to handle large-scale tables programmatically:

1. **Defensive Data Ingestion:** Raw datasets are loaded with all column types forced to characters (`col_types = cols(.default = "c")`). This operational constraint prevents incorrect automated type guessing by `readr` due to missing data flags or genomic non-numeric markers, ensuring high pipeline stability.

2. **Explicit Coercion:** Target demographic and variant indicators—specifically Allele Count (AC), Allele Number (AN), and Homozygote Count—are explicitly coerced to numeric fields.

3. **Long-Format Reshaping:** A programmatic loop iterates through each of the 10 target sub-populations. For each population, it extracts the relevant columns, computes the

population-specific Allele Frequency (AF) as defined by Equation 2, assigns a structural categorical string label to a new `Population` column, and isolates these observations into a tidy data frame.

$$AF = \frac{AC}{AN} \quad \text{where } AN > 0 \tag{2}$$

4. **Aggregation:** The extracted individual population rows and the computed "Overall" global metrics are vertically stacked using row-binding operations. Rows containing missing data across both primary metrics (AC and AN) are removed.

5. **Consolidation:** The pipeline iterates across all three target genes, saves individual tidy data assets, and exports a master consolidated matrix (`all_genes_tidy.csv`) for structural analysis.

6. **Exploratory Cross-Tabulation Summary:** To evaluate the success of the data transformation and gain an initial understanding of variant density, a cross-tabulated matrix was compiled at the termination of the execution block. By applying a `pivot_wider` transformation on the aggregated metrics, individual variant counts were summarized across populations as a matrix format for descriptive transparency.

3 Descriptive Statistics and Exploratory Visualization Framework

Following the consolidation of the data pipeline, an Exploratory Data Analysis (EDA) framework was executed to discover the underlying statistical properties of the allele frequency (AF) distributions across the three target loci (*APOE*, *CYP2C19*, and *HLA-B*).

A defining characteristic of genomic variant datasets is their extreme zero-skewness, where an overwhelming majority of mutations reside at near-zero frequencies, while only a select few exhibit high-frequency penetrance. To capture these characteristics without sacrificing structural resolution, descriptive metrics were computed across diverse quantiles, and multi-panel visualization dashboards were programmatically rendered using `ggplot2` and `gridExtra`.

3.1 Quantile Summarization and Empirical Variance Profiles

To quantitatively evaluate the continuous allele frequency (AF) profiles across the target loci, extensive descriptive statistics and quantile boundaries were compiled. Because the empirical distributions are non-Gaussian and heavily right-skewed, standard metrics like the arithmetic mean (μ) and standard deviation (σ) fail to represent the center of mass accurately. Table 2 details the global summary metrics across all 5,328 pooled variant observations.

Table 2: Overall Descriptive Summary Statistics of Allele Frequencies (AF) across Target Loci

Gene	Count	Mean	Std. Dev.	Min	25%	Median (50%)	75%	Max
APOE	1421	0.000623	0.0176	6.195×10^{-7}	6.393×10^{-7}	1.240×10^{-6}	3.136×10^{-6}	0.6409
CYP2C19	2225	0.001175	0.0288	6.195×10^{-7}	6.200×10^{-7}	1.240×10^{-6}	3.721×10^{-6}	0.9387
HLA-B	1682	0.010895	0.0583	6.296×10^{-7}	9.723×10^{-7}	2.183×10^{-6}	9.427×10^{-6}	0.7803

To isolate the behavior of the distribution tails where low-frequency variations reside, a detailed high-resolution percentile analysis was constructed (Table 3).

Table 3: Detailed Percentile Profile of Continuous Allele Frequency (AF) Distributions

Gene	1%	5%	25%	Median (50%)	75%	95%	99%
APOE	6.195×10^{-7}	6.196×10^{-7}	6.393×10^{-7}	1.240×10^{-6}	3.136×10^{-6}	2.857×10^{-5}	2.899×10^{-4}
CYP2C19	6.195×10^{-7}	6.195×10^{-7}	6.200×10^{-7}	1.240×10^{-6}	3.721×10^{-6}	4.399×10^{-5}	7.206×10^{-4}
HLA-B	6.335×10^{-7}	7.114×10^{-7}	9.723×10^{-7}	2.183×10^{-6}	9.427×10^{-6}	0.0431	0.3223

The empirical discrepancies between the mean and median metrics highlight a severe right-skew. For example, in *CYP2C19*, the 75th percentile (3.721×10^{-6}) remains five orders of magnitude smaller than the absolute maximum observed value (0.9387). This structural pattern shifts noticeably within *HLA-B*, where the 95th (0.0431) and 99th (0.3223) percentiles confirm a substantially broader presence of intermediate and common polymorphisms.

3.2 Demographic Stratification of Variant Frequencies

To capture localized ancestral shifts, Table 4 presents the metrics stratified across the 10 separate ancestral sub-populations included in the gnomAD corpus.

A noticeable artifact of small-sample demographic groupings is observed where the median (50%) value falls directly to 0.0 across almost all individual cohorts. This occurs because highly localized rare variants frequently record an allele count of zero within constrained sub-samples. Conversely, the *European (non-Finnish)* demographic (which comprises the largest sample allocation) exhibits non-zero median values across all loci (e.g., 1.431×10^{-6} for *HLA-B*), illustrating how sample size fluctuations impact empirical zero-mass estimates.

Table 4: Stratified Summary Statistics of Allele Frequencies (AF) by Ancestral Sub-Population

Gene	Ancestral Sub-Population	Count	Mean	Std. Dev.	Median (50%)	Max
APOE	Admixed American	1421	5.104×10^{-4}	0.0152	0.0	0.5619
	African/African American	1421	8.856×10^{-4}	0.0237	0.0	0.8596
	Amish	1421	5.944×10^{-4}	0.0188	0.0	0.6963
	Ashkenazi Jewish	1421	5.936×10^{-4}	0.0170	0.0	0.6267
	East Asian	1421	4.271×10^{-4}	0.0113	0.0	0.4114
	European (Finnish)	1421	6.912×10^{-4}	0.0200	0.0	0.7292
	European (non-Finnish)	1421	6.233×10^{-4}	0.0176	8.621×10^{-7}	0.6418
	Middle Eastern	1421	5.124×10^{-4}	0.0149	0.0	0.5527
	Remaining	1421	5.994×10^{-4}	0.0168	0.0	0.6153
	South Asian	1421	5.027×10^{-4}	0.0142	0.0	0.5230
CYP2C19	Admixed American	2225	0.0011	0.0291	0.0	0.9600
	African/African American	2225	0.0013	0.0292	0.0	0.9875
	Amish	2225	0.0012	0.0301	0.0	0.9704
	Ashkenazi Jewish	2225	0.0011	0.0280	0.0	0.9183
	East Asian	2225	0.0015	0.0305	0.0	0.9601
	European (Finnish)	2225	0.0012	0.0292	0.0	0.9440
	European (non-Finnish)	2225	0.0011	0.0287	8.480×10^{-7}	0.9377
	Middle Eastern	2225	0.0011	0.0281	0.0	0.9276
	Remaining	2225	0.0012	0.0288	0.0	0.9388
	South Asian	2225	0.0014	0.0299	0.0	0.8894
HLA-B	Admixed American	1682	0.0133	0.0701	0.0	0.8923
	African/African American	1682	0.0106	0.0549	0.0	0.7942
	Amish	1682	0.0103	0.0617	0.0	0.7812
	Ashkenazi Jewish	1682	0.0156	0.0801	0.0	0.9396
	East Asian	1682	0.0140	0.0722	0.0	0.8178
	European (Finnish)	1682	0.0091	0.0512	0.0	0.7289
	European (non-Finnish)	1682	0.0102	0.0562	1.431×10^{-6}	0.7569
	Middle Eastern	1682	0.0148	0.0748	0.0	0.9280
	Remaining	1682	0.0116	0.0611	0.0	0.8059
	South Asian	1682	0.0161	0.0812	0.0	0.9333

3.3 Overall Allele Frequency Distribution

As a first step, the allele frequency distributions of the three genes were examined across all variants using the overall population data from gnomAD. Figure 1 displays the distributions on both linear and log scale. On the linear scale, the vast majority of variants are compressed near zero, making it impossible to distinguish among them. The log-scale subplots reveal the true shape of the distribution: a steeply declining spectrum, with most variants concentrated between 10^{-6} and 10^{-5}, and very few reaching frequencies above 10^{-2}. This L-shaped pattern is the hallmark of the site frequency spectrum.

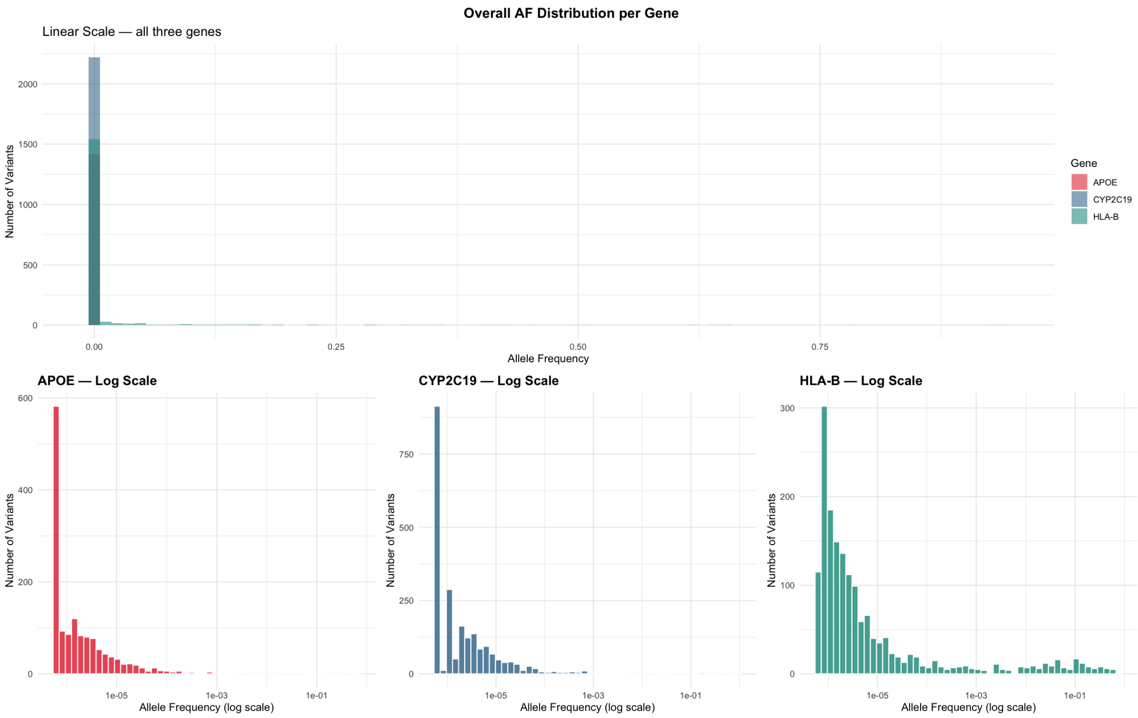


Figure 1: Empirical distribution of Allele Frequencies (AF) plotted on linear and base-10 logarithmic scales to expose low-frequency architectural variations.

3.4 Stratified Comparative Analysis by Ancestral Sub-Population

To assess how variant characteristics fluctuate across localized demographic boundaries, the categorical population indicators were cross-examined. Populations are sorted dynamically along the horizontal axis in descending order based on their true median allele frequency.

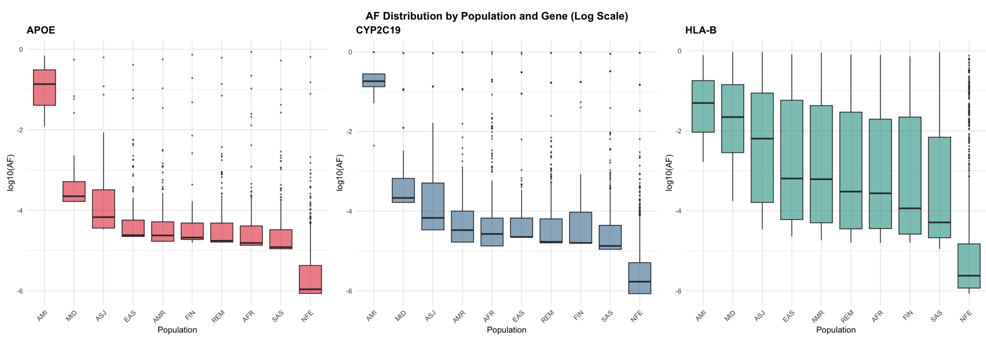


Figure 2: Log-transformed boxplot analysis of allele frequencies across 10 distinct ancestral populations, ordered descending by population-specific median frequencies.

This sorting paradigm eliminates uninformative layout noise. As diagrammed in Figure 2, plotting $\log_{10}(AF)$ across these ordered cohorts provides immediate visibility into which genetic ancestors serve as primary reservoirs for variant variance.

3.5 Discrete Frequency Class and Bracketing Metrics

The final exploratory framework transitions from continuous density estimation to discrete categorical partitioning. The continuous sample space of allele frequencies was mapped into five standard population-genetics intervals using interval splitting thresholds:

$$\mathbf{B} = \{0,\ 0.001,\ 0.01,\ 0.05,\ 0.5,\ 1.0\} \tag{3}$$

These boundaries divide the variants into standard classes: Ultra-rare ($< 0.1\%$), Rare (0.1%–1%), Intermediate (1%–5%), Common (5%–50%), and High-frequency ($> 50\%$).

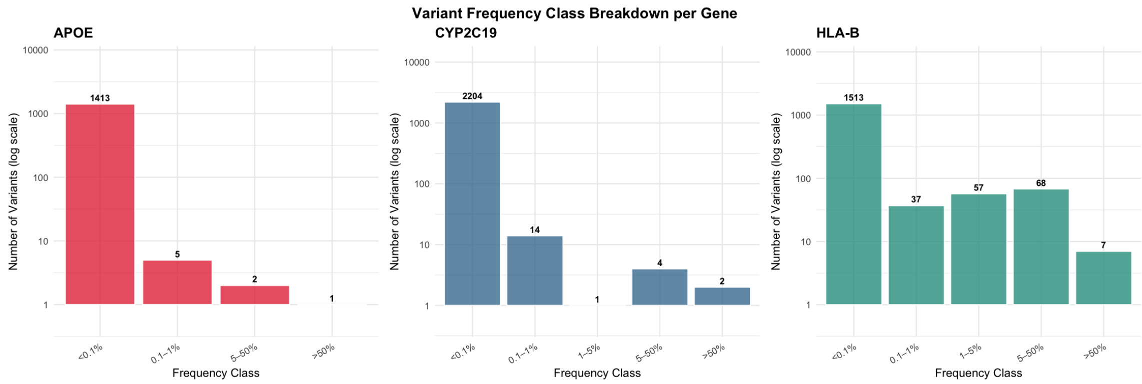


Figure 3: Discrete frequency class breakdown across five standardized operational intervals, demonstrating the numeric dominance of low-frequency alleles.

Because the frequency classes contain highly disparate numeric quantities—where ultra-rare variations outnumber high-frequency variants by multiple orders of magnitude—a standard linear scale would compress the high-frequency columns into complete invisibility. To address this scaling disparity, the vertical axis utilizes an explicit `scale_y_log10` scaling transformation.

3.6 Empirical Cumulative Density Functions (ECDF)

The empirical cumulative distribution function (ECDF) of allele frequency provides a clear visual summary of rare variant prevalence. The empirical estimator $F_n(x)$ is defined as the fraction of observations less than or equal to a target allele frequency threshold:

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(AF_i \leq x) \tag{4}$$

where $\mathbb{I}$ represents the indicator function. Figure 4 displays the log-scaled continuous ECDF curves overlaid with standard operational genetics boundaries at 0.1%, 1%, and 5% thresholds.

The geometric slopes of the ECDF tracks yield direct insight into mutational densities. For *APOE* and *CYP2C19*, the cumulative tracks rise with extreme velocity, intersecting the 1% dashed marker ($x = 10^{-2}$) well above the 95% boundary. This confirms an immense concentration of low-frequency variants. Conversely, the curve for *HLA-B* exhibits a significantly suppressed trajectory, establishing that a large proportion of its variants migrate into intermediate and high-frequency domains.

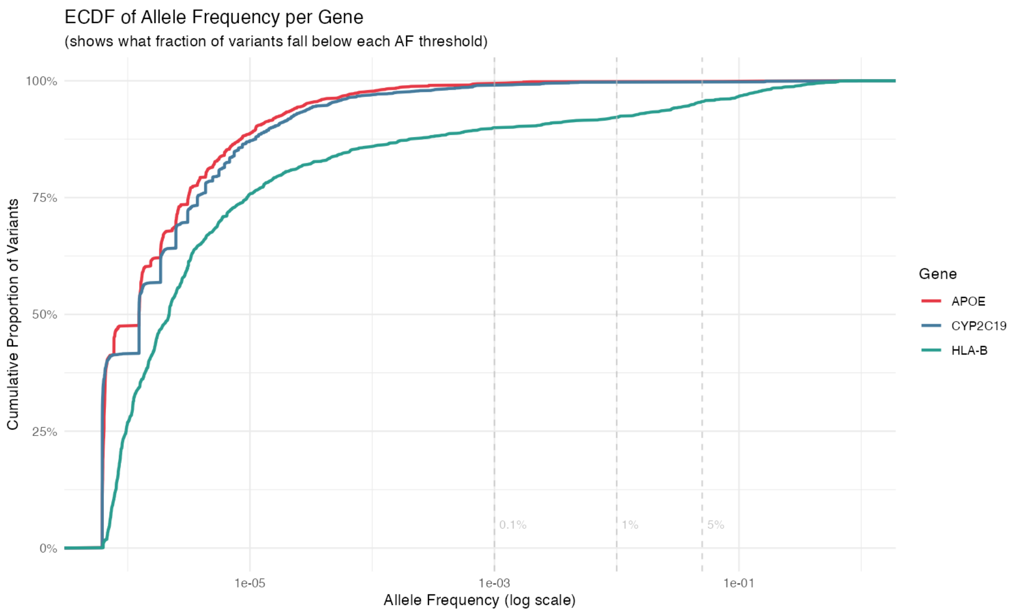


Figure 4: Empirical Cumulative Distribution Function (ECDF) of allele frequencies across target genes, demonstrating distinct tail-density accumulations.

3.7 Cross-Demographic Covariance Matrix Heatmaps

To examine population-level variation, the median $\log_{10}(\mathrm{AF})$ was computed for each gene–population pair and visualised as a heatmap (Figure 5). Two patterns stand out. First, the Amish population (AMI) consistently shows the highest median AF across all three genes (values of -0.9, -0.7, and -1.3 for APOE, CYP2C19, and HLA-B respectively). This is a well-documented consequence of the Amish founder effect: a small number of 18th-century founders gave rise to the entire modern Amish population, causing certain variants to drift to much higher frequencies than in outbred populations. Second, the non-Finnish European population (NFE) shows the lowest median AF across all genes (values of -6.0, -5.8, and -5.6). This is attributable to NFE being the largest cohort in gnomAD, giving it greater statistical power to detect variants at extremely low frequencies and thus pulling the median downward.

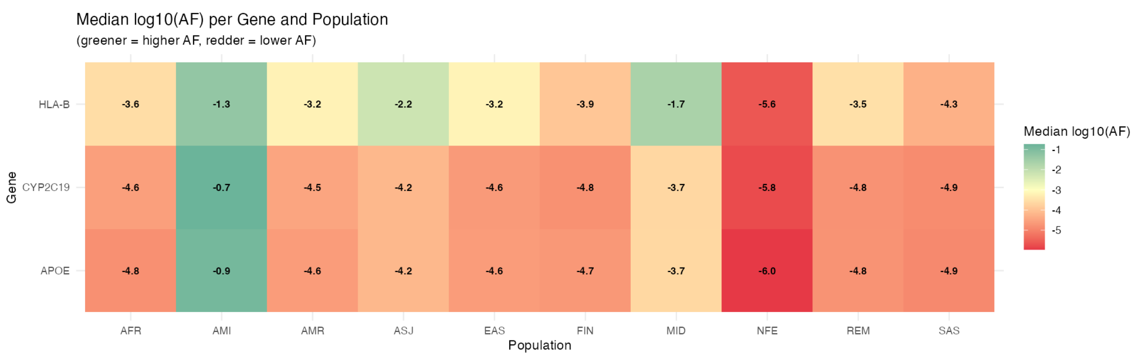


Figure 5: 2D Heatmap Matrix of median $\log_{10}(AF)$ values stratified by locus and ancestral sub-population.

3.8 Invariance of Rare Variant Proportions

Figure 6 shows the proportion of variants with $AF < 1\%$ for each gene across all ten populations. APOE and CYP2C19 reach 100% in virtually every population, while HLA-B consistently falls between 91% and 95%. The uniformity across populations for APOE and CYP2C19 suggests that the rarity of variants in these genes is a global phenomenon driven by functional constraint, rather than a population-specific effect.

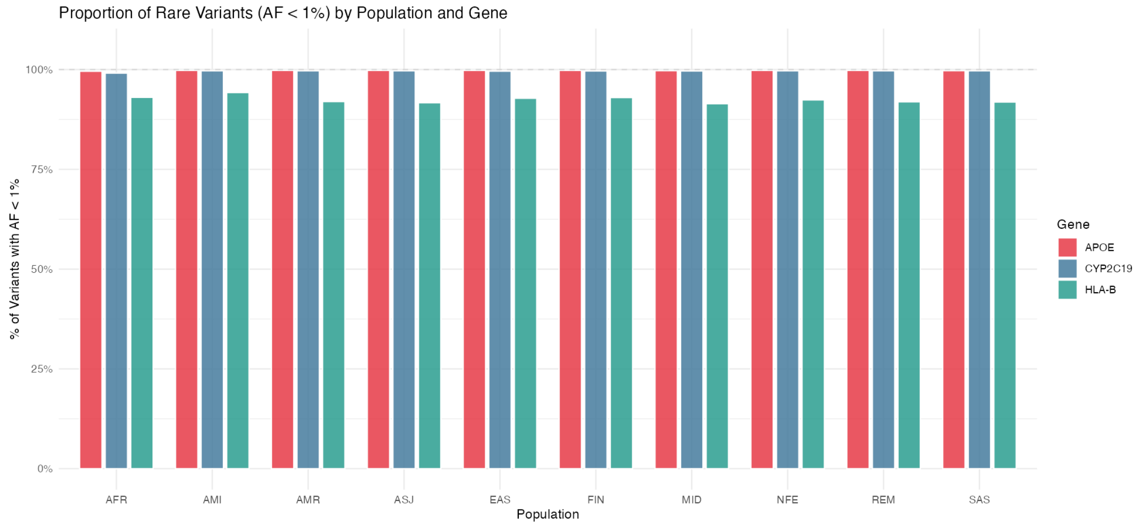


Figure 6: Grouped bar chart illustrating the percentage distribution of rare variants ($AF < 1\%$) across independent ancestral categories.

4 Modeling

4.1 Distribution Selection and Theoretical Motivation

The goal of the modeling step is to fit a parametric probability distribution that faithfully describes the allele frequency (AF) spectrum observed in the data. Since allele frequencies are by definition bounded to the interval $[0, 1]$, only distributions defined on this support are natural candidates. The flexibility of the Beta distribution across such different shapes makes it a natural first choice for modelling AF spectra, which can range from strongly J-shaped (APOE, CYP2C19) to moderately bell-shaped (HLA-B under balancing selection). Figure 7 illustrates these regimes.

4.2 The Beta Distribution

The **Beta distribution**, $\text{Beta}(\alpha, \beta)$, is a two-parameter family of continuous probability distributions defined on $[0, 1]$. Its probability density function is:

$$f(x \mid \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \qquad x \in [0, 1], \ \alpha > 0, \ \beta > 0 \tag{5}$$

where $\mathrm{B}(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function acting as a normalisation constant. The two shape parameters α and β jointly determine the shape of the distribution:

- $\alpha < 1, \ \beta > 1$: **J-shaped** — density concentrated near zero, describing a population dominated by ultra-rare variants.
- $\alpha < 1, \ \beta < 1$: **U-shaped** — two peaks at 0 and 1, describing a bimodal spectrum where variants are either very rare or very common.
- $\alpha > 1, \ \beta > 1$: **bell-shaped** — interior mode, typical of intermediate-frequency variants.
- $\alpha = \beta = 1$: **Uniform** — all frequencies equally likely.

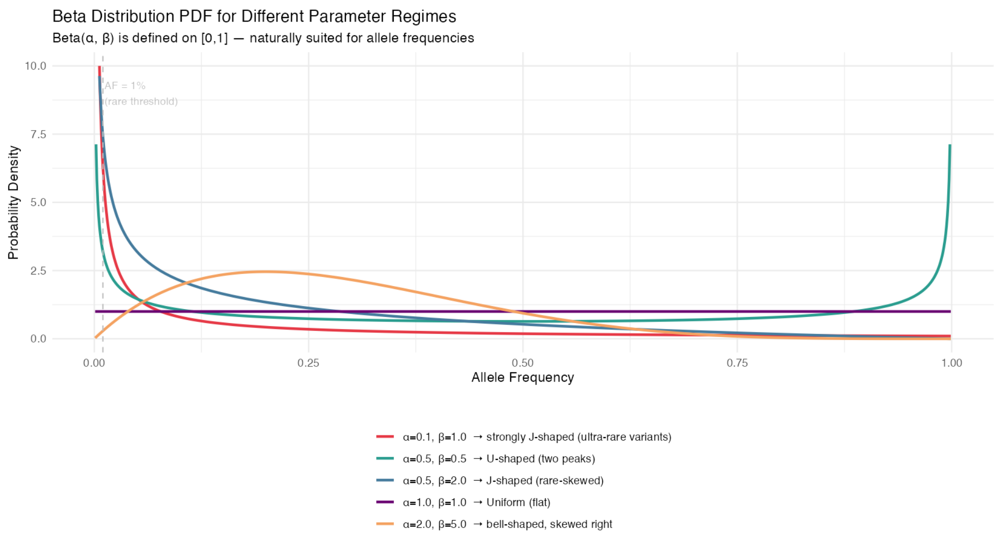


Figure 7: Theoretical probability density functions of the Beta distribution across diverse parameter regimes, demonstrating extreme geometric flexibility.

4.3 Biological Justification: The Wright-Fisher Stationary State

Beyond its mathematical convenience, the Beta distribution has a direct theoretical justification in population genetics. Under the **Wright–Fisher model** of genetic drift with mutation, the stationary distribution of allele frequencies at a locus in a diploid population of effective size N_e and per-generation mutation rate μ converges to:

$$p(x) \propto x^{\theta-1}(1-x)^{\theta-1}, \qquad \theta = 4N_e\mu \tag{6}$$

which is precisely Beta(θ, θ). For typical human populations ($N_e \approx 10{,}000$, $\mu \approx 10^{-5}$), $\theta = 4 \times 10^4 \times 10^{-5} = 0.4$, giving a strongly J-shaped distribution consistent with the ultra-rare variant excess observed in APOE and CYP2C19. For HLA-B, where balancing selection elevates effective mutation rate, a larger $\theta \approx 2.0$ produces the flatter, bell-shaped curve compatible with the observed broader frequency spectrum. This is shown in Figure 8.

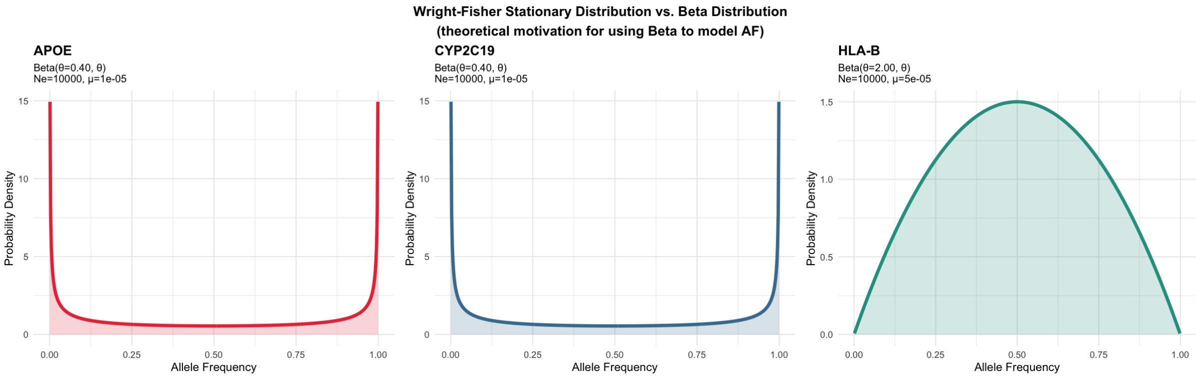


Figure 8: Wright–Fisher stationary Beta distributions for biologically motivated parameters. APOE and CYP2C19 use $\theta = 0.40$ ($N_e = 10{,}000$, $\mu = 10^{-5}$), giving a strongly J-shaped curve. HLA-B uses $\theta = 2.00$ ($\mu = 5 \times 10^{-5}$, reflecting balancing selection), giving a bell-shaped curve consistent with its broader allele frequency spectrum.

4.4 Parameter Estimation via Maximum Likelihood

4.4.1 Method

For each gene, the parameters of the Beta and Log-normal distributions were estimated by **Maximum Likelihood Estimation** (MLE) using the overall population allele frequency data.

For the **Beta distribution**, given a sample $x_1, \ldots, x_n \in (0,1)$, MLE maximises the log-likelihood:

$$\ell(\alpha, \beta) = (\alpha - 1)\sum_{i=1}^{n} \ln x_i + (\beta - 1)\sum_{i=1}^{n} \ln(1 - x_i) - n \ln B(\alpha, \beta) \tag{7}$$

with respect to $\alpha > 0$ and $\beta > 0$. In R, this was performed using `fitdist(af, "beta", method = "mle")` from the `fitdistrplus` package, which returns the estimates $\hat{\alpha}$ and $\hat{\beta}$ together with their standard errors, AIC, and BIC. No location or scale constraints are needed because R's Beta distribution is parameterised natively on $[0, 1]$.

For the **Log-normal distribution**, MLE has a closed-form solution:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} \ln x_i, \qquad \hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\ln x_i - \hat{\mu})^2} \tag{8}$$

estimated in R via `fitdist(af, "lnorm", method = "mle")`. Note that $\hat{\mu}$ and $\hat{\sigma}$ here are the mean and standard deviation of ln(AF) on the natural log scale, not the mean and standard deviation of AF itself.

Both fits were performed on variants with AF $\in (0, 1)$ (boundary values excluded, as the Beta density is undefined at 0 and 1 when $\alpha < 1$ or $\beta < 1$).

4.4.2 Results

Table 5 reports the MLE estimates for both distributions across the three genes. Figure 9 overlays the fitted PDFs on the observed $\log_{10}$(AF) histograms.

Table 5: MLE parameter estimates for the Beta and Log-normal distributions fitted to the overall allele frequency data of each gene. $\hat{\mu}$ and $\hat{\sigma}$ for the Log-normal are on the natural log scale.

Gene	Beta MLE		Log-normal MLE	
	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\mu}$	$\hat{\sigma}$
APOE	0.127	140.638	−13.22	1.46
CYP2C19	0.111	40.219	−13.08	1.63
HLA-B	0.107	8.228	−11.83	3.24

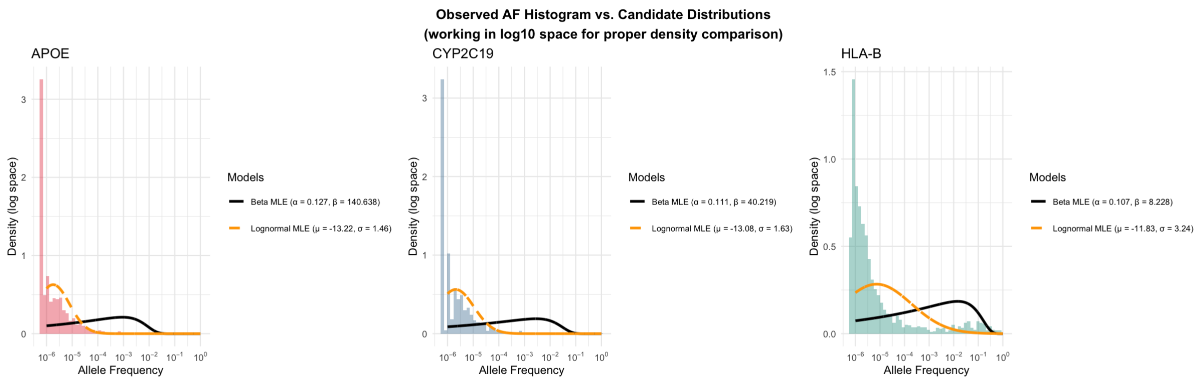


Figure 9: Observed AF histograms (coloured bars, in $\log_{10}$ space) overlaid with Beta MLE (solid black) and Log-normal MLE (dashed orange) fits. Neither distribution perfectly captures the sharp ultra-rare spike, but both approximate the bulk of the spectrum. The x-axis shows actual allele frequency values on a log scale; density is expressed in $\log_{10}$ space using the appropriate Jacobian transformation.

4.4.3 Interpretation of the Overall Fits

Several patterns are immediately apparent from Table 5 and Figure 9.

$\hat{\alpha}$ is small and similar across all three genes. All three Beta fits yield $\hat{\alpha} < 0.13$, confirming that the distribution is strongly J-shaped in every gene: probability density is highest near AF $= 0$ and decays monotonically. This is the universal signature of purifying selection keeping most variants rare, and it is consistent across genes despite their very different biological functions.

$\hat{\beta}$ encodes gene-specific rarity. The estimated $\hat{\beta}$ differs dramatically: 140.6 for APOE, 40.2 for CYP2C19, and only 8.2 for HLA-B. Since the mean of a Beta distribution is $\mu = \hat{\alpha}/(\hat{\alpha} + \hat{\beta})$, a larger $\hat{\beta}$ with fixed small $\hat{\alpha}$ implies a smaller mean AF. Concretely: $\bar{\mathrm{AF}}_{\mathrm{APOE}} \approx 9 \times 10^{-4}$, $\bar{\mathrm{AF}}_{\mathrm{CYP2C19}} \approx 2.8 \times 10^{-3}$, and $\bar{\mathrm{AF}}_{\mathrm{HLA-B}} \approx 1.3 \times 10^{-2}$. HLA-B's much lower $\hat{\beta}$ reflects the action of balancing selection, which prevents variants from remaining ultra-rare and produces a substantially broader frequency spectrum.

The Log-normal $\hat{\sigma}$ reflects spectral breadth. The estimated $\hat{\sigma}$ increases from 1.46 (APOE) to 1.63 (CYP2C19) to 3.24 (HLA-B). A larger $\hat{\sigma}$ means the distribution of ln(AF) is more spread out — more variants at both very low and moderately high frequencies. HLA-B's $\hat{\sigma} = 3.24$ is more than twice that of APOE, again reflecting the broader frequency spectrum maintained by balancing selection.

Visual fit quality. Inspecting Figure 9, neither model perfectly reproduces the observed histograms. The Beta MLE curve (solid black) peaks consistently to the right of the histogram mode, underestimating the ultra-rare spike at AF $\lesssim 10^{-5}$ and overestimating density at intermediate frequencies. The Log-normal MLE curve (dashed orange) tracks the bulk of the distribution more closely for APOE and CYP2C19 in log space, but also fails to capture the sharp leftmost spike. For HLA-B, neither fit is satisfactory: the broader spectrum with a long moderate-frequency tail exceeds what either two-parameter model can describe.

This visual evidence motivates the formal goodness-of-fit evaluation in Section 5, and suggests that the structural limitation is not a failure of the estimation procedure but an intrinsic constraint of fitting unimodal two-parameter distributions to a bimodal AF spectrum.

4.5 Stratified Parametric Modeling Across Ancestral Cohorts

4.5.1 Methodology

To determine if the shape parameters identified in Section 4.4 represent universal gene-specific architectures or are heavily modulated by ancestral population backgrounds, the maximum likelihood estimation pipeline was stratified. The continuous dataset was partitioned into 10 distinct non-overlapping sub-populations as defined by the gnomAD census metadata.

For each individual gene-population pair ($j \in \{1, \ldots, 30\}$), independent optimization sweeps were executed. To protect the numeric optimizer against boundary instability and small-sample distortions, a defensive filtering constraint was enforced:

$$\mathbf{x}_{jk} = \{AF \in \mathbf{x}_j \mid 0 < AF < 1\} \tag{9}$$

A sub-population was successfully advanced to the numerical solver if and only if the absolute variant sample size satisfied $n \geq 10$. Configurations falling below this threshold were classified as non-convergent to prevent the propagation of high-variance, unstable parameter boundaries.

4.5.2 Results and Parameter Matrix Topology

The stratified optimization successfully converged for 28 out of the 30 target demographic matrices. For the *Amish* cohort (`AMI`), the filtered variant array sizes fell below the operational sample constraint for both *APOE* ($n = 0$) and *CYP2C19* ($n = 0$), reflecting the severe founder-effect genetic bottlenecks characteristic of this isolated lineage.

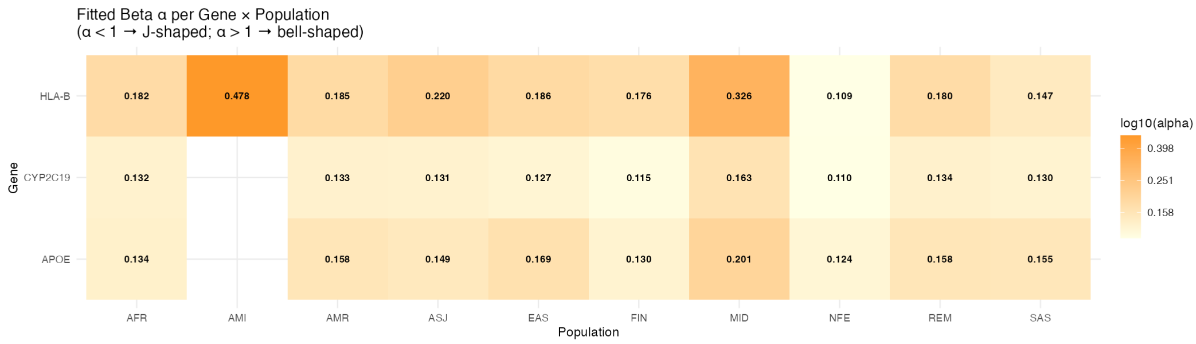


Figure 10: Log-scaled heatmap matrix of the fitted Beta α shape parameters across individual genetic loci and ancestral sub-populations.

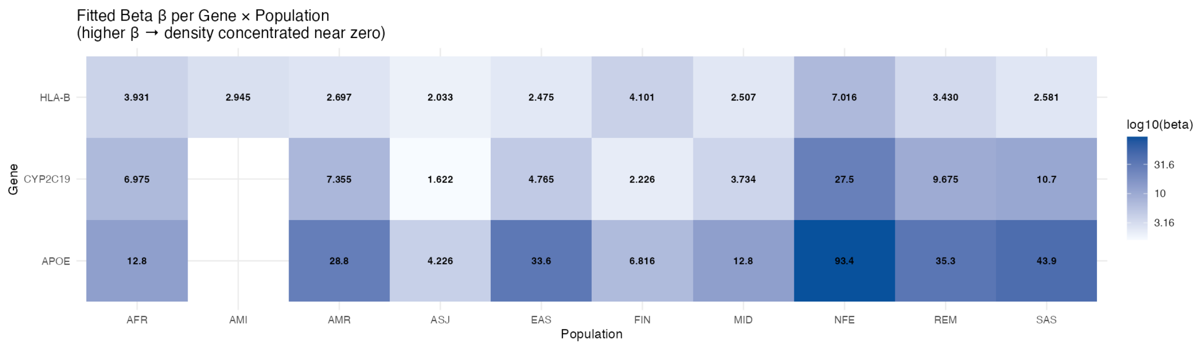


Figure 11: Log-scaled heatmap matrix of the fitted Beta β tail-weight parameters, illustrating extreme macro-level variations across loci.

The resulting spatial distribution of the fitted shape values is visualized via cross-tabulated logarithmic heatmaps for the alpha parameter (Figure 10) and the beta parameter (Figure 11).

The matrix landscapes reveal an explicit ordering of parameters. **The scale-shaping parameter α remains tightly constrained within a narrow, low-value window ($0.109 \leq \alpha \leq 0.478$) across all geographic contexts, demonstrating that the underlying distribution maintains its characteristic J-shape globally.**

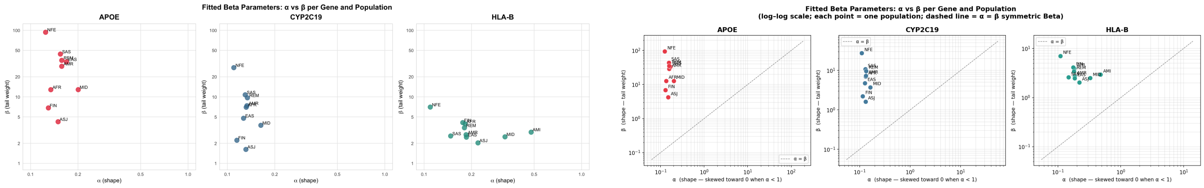
In sharp contrast, the tail-weight parameter β experiences multi-order fluctuations. The *European (non-Finnish)* cohort (`NFE`) represents a major statistical anchor point across all heatmaps, exhibiting extreme parameter inflation ($\alpha_{\text{APOE}} = 0.124, \beta_{\text{APOE}} = 93.4$). This pattern reflects the immense sample sizing and sequencing depth of the European repository, which captures a vast spectrum of unique singletons that drag the model's tail weight upward.

4.6 $\hat{\alpha}$ vs $\hat{\beta}$ Scatter

To analyze parameter covariance, we plotted our fitted α shape metrics against their corresponding β tail weights on a log-log scale. During this process, we encountered a major visual limitation when trying to generate the final figures natively in R.

As shown in Figure 12a, the plot generated in R via `ggplot2` fails to communicate the data clearly. Because our alpha values are restricted to a very narrow window ($0.10 \leq \alpha \leq 0.50$), R heavily warps the coordinate space. This warping distorts the grid lines, completely removing the vital 45-degree trajectory of the $\alpha = \beta$ diagonal reference line.

To resolve this issue, we exported our data coordinates to Python and generated the final subplots using `matplotlib` (Figure 12b). Python maintains the exact same numerical values but



(a) Scatter rendering in R (`ggplot2`). **(b)** Optimized rendering in Python (`matplotlib`).

Figure 12: Faceted log-log scatter plots of fitted α versus β parameters per gene and sub-population. While the underlying numerical data is identical, Python provides much better scale alignment and label spacing.

sets a uniform log-scale viewport starting cleanly at 10^{-1}. This fixes the geometric distortion, keeping the diagonal identity line at a true, readable 45-degree angle so the relative heights and positions of our genetic clusters are instantly clear.

4.7 Interpretation of Parameter Clusters

Using the clearer Python visualization (Figure 12b) we can see that the dashed diagonal (which marks the symmetric Beta ($\alpha = \beta$)), shows that all points lie well above it ($\hat{\beta} \gg \hat{\alpha}$), confirming that the distribution is strongly asymmetric toward zero in all cases. Aslo we can identify distinct evolutionary signatures for each gene across global populations:

- ***APOE*** **(Red Locus):** Characterized by very low α shapes and high tail weights ($\beta \geq 4.226$), sitting furthest above the diagonal parity line. This positions the gene in a highly conserved zone under aggressive purifying selection globally.
- ***CYP2C19*** **(Blue Locus):** Forms a transitional intermediary cluster with moderate functional constraints and balanced tail-decay weights.
- ***HLA-B*** **(The Concentrated Spot):** In contrast to APOE and CYP2C19, HLA-B shows a much tighter clustering of parameter estimates across populations. This pattern is consistent with balancing selection maintaining a more stable allele-frequency spectrum, although demographic history likely still contributes to the remaining variation.

Additionally, the European (`NFE`) cohort stands out as a massive vertical outlier in the *APOE* and *CYP2C19* panels ($\beta_{\text{APOE}} = 93.429$, $\beta_{\text{CYP2C19}} = 27.462$). This is a sample size effect rather than a unique biological mechanism; the massive size of the European census group ($n_{\text{NFE}} = 1527$) captures an enormous pool of rare singletons, which mathematically pulls the MLE tail-weight estimator upward relative to smaller cohorts.

5 Evaluation

This section assesses how well the fitted Beta distributions reproduce the observed allele frequency data. The analyses were performed in R: the visual overlays using `ggplot2` with the Jacobian-corrected Beta PDF, and the KS test using the `dgof` package, which handles ties and continuous distributions more robustly than base R's `ks.test`.

5.1 Visual Comparison: Observed AF vs. Fitted Beta

Figures 13–15 display, for each gene, a 2×5 grid of panels (one per population) where the coloured histogram shows the observed distribution of $\log_{10}(\mathrm{AF})$ and the black curve shows the fitted Beta PDF transformed into $\log_{10}$ space via the Jacobian:

$$f_{\log}(y) \;=\; f_{\mathrm{Beta}}\Big(10^{y};\, \hat{\alpha}, \hat{\beta}\Big) \cdot 10^{y} \cdot \ln 10 \tag{10}$$

where $y = \log_{10}(\mathrm{AF})$. This correction is necessary because the histogram bins are uniform in $\log_{10}$ space, not in linear AF space; without it the curve and the bars would be on incompatible density scales. In R, the transformation was implemented as `dbeta(x, a, b) * x * log(10)` evaluated on a fine grid `x = 10^y`.

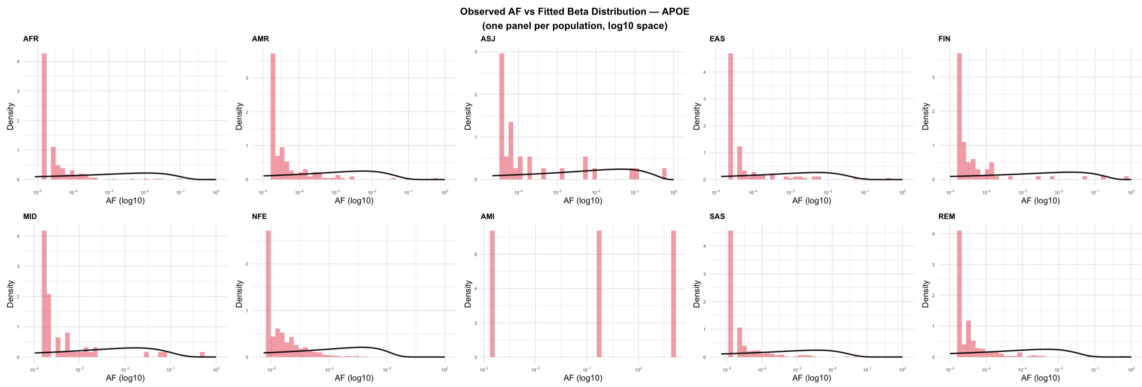


Figure 13: Observed $\log_{10}(\mathrm{AF})$ histograms (pink bars) overlaid with the Jacobian-corrected fitted Beta PDF (black curve) for *APOE*, one panel per population. In every panel the curve lies far below the histogram spike at $\mathrm{AF} \lesssim 10^{-4}$ and rises slowly toward intermediate frequencies, indicating that the Beta severely underestimates ultra-rare variants. The AMI panel contains insufficient data for MLE.

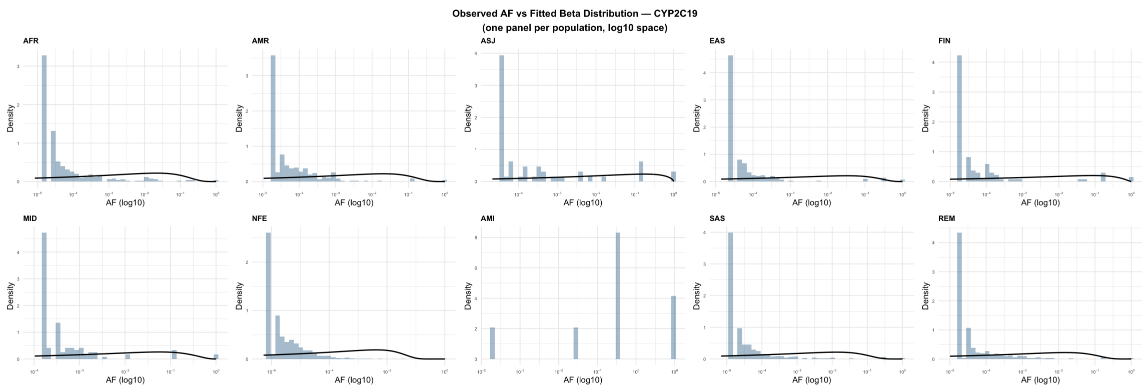


Figure 14: Same as Figure 13 for *CYP2C19* (blue). The pattern of systematic displacement is nearly identical to APOE across all nine populations, confirming that the misfit is a structural property of the AF spectrum rather than a gene- or population-specific artefact. The AMI panel is again insufficient.

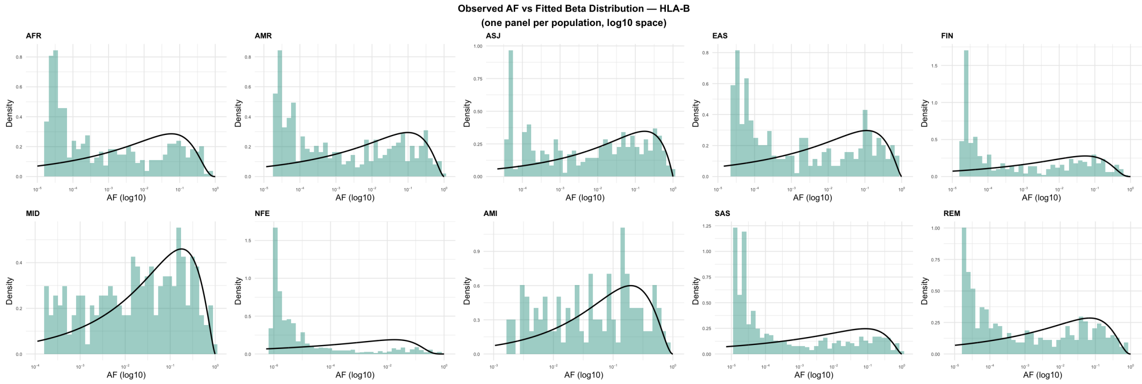


Figure 15: Same as Figure 13 for *HLA-B* (teal). The Beta curves track the histograms substantially better than for the other two genes. In particular, MID and AMI show the curve passing through the bulk of the histogram, while NFE retains the familiar displaced pattern driven by its large cohort size.

The visual inspection reveals a consistent and systematic pattern for APOE and CYP2C19: the fitted Beta PDF is displaced several orders of magnitude to the right of the observed histogram peak. Observed data peak at AF $\approx 10^{-5}$, while the Beta curve peaks near AF $\approx 10^{-2}$. This displacement cannot be corrected by further optimisation of the MLE: it reflects the fundamental limitation of a unimodal two-parameter distribution that must simultaneously accommodate both the ultra-rare spike and the long moderate-frequency tail. The MLE estimator compromises between the two regimes and fits neither perfectly.

HLA-B behaves differently. Its broader frequency spectrum (maintained by balancing selection) reduces the severity of the spike and allows the Beta curve to follow the general shape of several population-level histograms, particularly AMI and MID. The NFE panel is the outlier within HLA-B: the very large cohort ($n = 1{,}294$) detects an unusually high number of singletons and doubletons that recreate the ultra-rare spike, pulling the histogram away from the curve.

5.2 Formal Goodness-of-Fit: Kolmogorov–Smirnov Test

5.2.1 Method

The **one-sample Kolmogorov–Smirnov (KS) test** provides a formal statistical assessment of how well a fitted distribution describes observed data. Given a sample $x_1, \ldots, x_n$ and a hypothesised CDF F_0, the KS statistic is:

$$D_n = \sup_x \left| F_n(x) - F_0(x) \right| \tag{11}$$

where F_n is the empirical CDF. The null hypothesis H_0 states that the data follow F_0; large D_n and small p-values lead to rejection of H_0.

Here F_0 is the Beta CDF with MLE-fitted parameters $(\hat{\alpha}, \hat{\beta})$ for each gene–population pair. The test was performed in R using `dgof::ks.test(af, "pbeta", alpha, beta)`, which passes the theoretical CDF directly as a string alongside its parameters a clean one-line call. An important caveat applies: because $\hat{\alpha}$ and $\hat{\beta}$ are estimated from the same data used in the test, the KS p-values are slightly anti-conservative (true critical values are larger than standard ones). The D_n statistic itself is unaffected by this and remains a valid measure of maximum CDF discrepancy.

5.2.2 Results

Tables 6–8 report the full KS results. Figure 16 summarises the D_n values as a gene × population heatmap.

Table 6: KS test results for *APOE*. All nine populations reject H_0 at $\alpha = 0.05$. D_n ranges from 0.374 (ASJ) to 0.484 (FIN).

Pop	n	$\hat{\alpha}$	$\hat{\beta}$	D_n	p**-value**
AFR	221	0.134	12.810	0.458	$< 10^{-15}$
AMR	179	0.158	28.769	0.416	$< 10^{-15}$
ASJ	31	0.149	4.226	0.374	2.10×10^{-4}
EAS	144	0.169	33.626	0.407	$< 10^{-15}$
FIN	78	0.130	6.816	0.484	3.33×10^{-16}
MID	61	0.201	12.826	0.408	2.89×10^{-9}
NFE	961	0.124	93.429	0.435	$< 10^{-15}$
SAS	264	0.155	43.940	0.429	$< 10^{-15}$
REM	267	0.158	35.337	0.439	$< 10^{-15}$

Table 7: KS test results for *CYP2C19*. All nine populations reject H_0. D_n ranges from 0.360 (ASJ) to 0.465 (EAS).

Pop	n	$\hat{\alpha}$	$\hat{\beta}$	D_n	p**-value**
AFR	384	0.132	6.975	0.404	$< 10^{-15}$
AMR	357	0.133	7.355	0.417	$< 10^{-15}$
ASJ	53	0.131	1.622	0.360	2.16×10^{-6}
EAS	241	0.127	4.765	0.465	$< 10^{-15}$
FIN	105	0.115	2.226	0.462	$< 10^{-15}$
MID	116	0.163	3.734	0.436	$< 10^{-15}$
NFE	1527	0.110	27.462	0.448	$< 10^{-15}$
SAS	480	0.130	10.731	0.437	$< 10^{-15}$
REM	428	0.134	9.675	0.418	$< 10^{-15}$

Table 8: KS test results for *HLA-B*. Nine of ten populations reject H_0; AMI is the only non-rejected fit ($D_n = 0.105$, $p = 0.111$). D_n values are substantially lower than for APOE and CYP2C19.

Pop	n	$\hat{\alpha}$	$\hat{\beta}$	D_n	p**-value**
AFR	425	0.182	3.931	0.203	1.33×10^{-15}
AMR	384	0.185	2.697	0.184	1.03×10^{-11}
ASJ	294	0.220	2.033	0.130	9.69×10^{-5}
EAS	356	0.186	2.475	0.192	8.28×10^{-12}
FIN	394	0.176	4.101	0.226	$< 10^{-15}$
MID	232	0.326	2.507	0.098	2.22×10^{-2}
NFE	1294	0.109	7.016	0.374	$< 10^{-15}$
AMI	130	0.478	2.945	0.105	0.111 **(fail to reject)**
SAS	578	0.147	2.581	0.265	$< 10^{-15}$
REM	421	0.180	3.430	0.202	2.55×10^{-15}

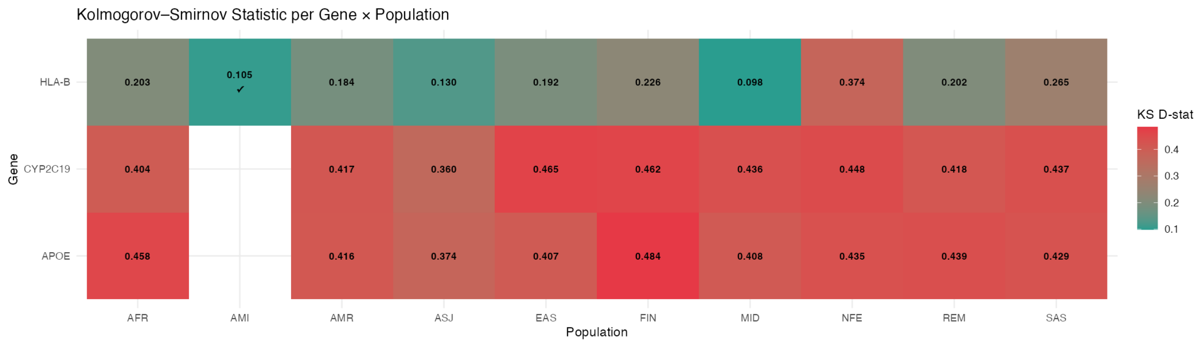


Figure 16: Heatmap of the KS D_n statistic per gene and population, produced with `ggplot2::geom_tile`. Teal indicates a small discrepancy (good fit); red indicates a large discrepancy (poor fit). APOE and CYP2C19 are uniformly deep red ($D_n \approx 0.37$–0.48). HLA-B shows a gradient from teal-green (AMI, MID, ASJ) to red (NFE). The tick mark (✓) in the HLA-B AMI cell marks the only non-rejected fit.

The KS results are unambiguous: **27 of 28 converged fits reject** H_0 at the $\alpha = 0.05$ level, the majority with p-values effectively equal to zero. For APOE and CYP2C19 the mean D_n is ≈ 0.428, meaning the empirical and theoretical CDFs diverge by up to 43% at some point a large misfit by any standard. HLA-B achieves a substantially lower mean $D_n = 0.198$, roughly half that of the other two genes.

The **sole non-rejected fit** is HLA-B in the Amish population ($D_n = 0.105$, $p = 0.111$). This is the one setting where the Beta distribution cannot be statistically distinguished from the observed data: the Amish founder effect produces a reduced, less spike-dominated variant spectrum that is genuinely close to a Beta shape. Note also that HLA-B MID has the lowest D_n overall ($D_n = 0.098$) but is still rejected ($p = 0.022$); it narrowly crosses the threshold because its sample size ($n = 232$) gives the test sufficient power to detect the small remaining deviation.

5.2.3 QQ Plot Analysis

Figures 17–19 show quantile-quantile (QQ) plots for each gene. Each panel plots the theoretical Beta quantiles (x-axis) against the observed AF quantiles (y-axis) at equally spaced probability levels $p_i = i/(n+1)$. Theoretical quantiles were computed in R as `qbeta(probs, alpha, beta)`. If the Beta model fits perfectly, all points fall on the dashed diagonal $y = x$.

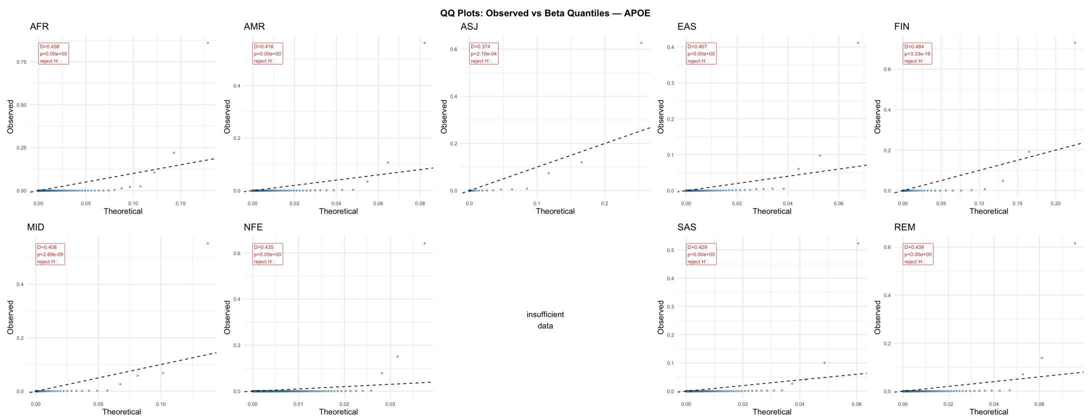


Figure 17: QQ plots for *APOE*, one panel per population. All panels show the same L-shaped collapse: nearly all observed quantiles are essentially zero while the theoretical Beta assigns them positive values up to ~ 0.2, then a handful of high-frequency outliers shoot far above the diagonal. This is the visual signature of a distribution that is too smooth and rightward-shifted to capture the observed ultra-rare spike.

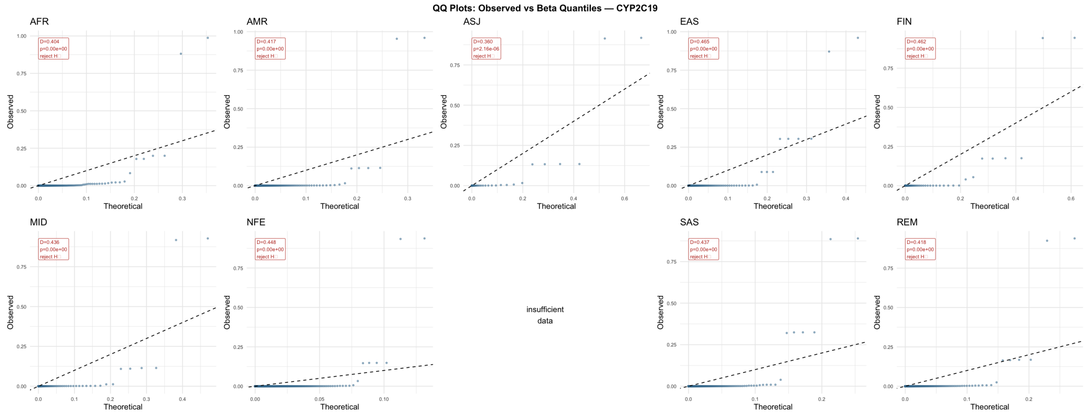


Figure 18: QQ plots for *CYP2C19*. The pattern is identical to APOE but more extreme: observed quantiles collapse along the x-axis for virtually all probability levels, with isolated high-frequency points far above the diagonal. The same structural failure of the Beta model appears in every population.

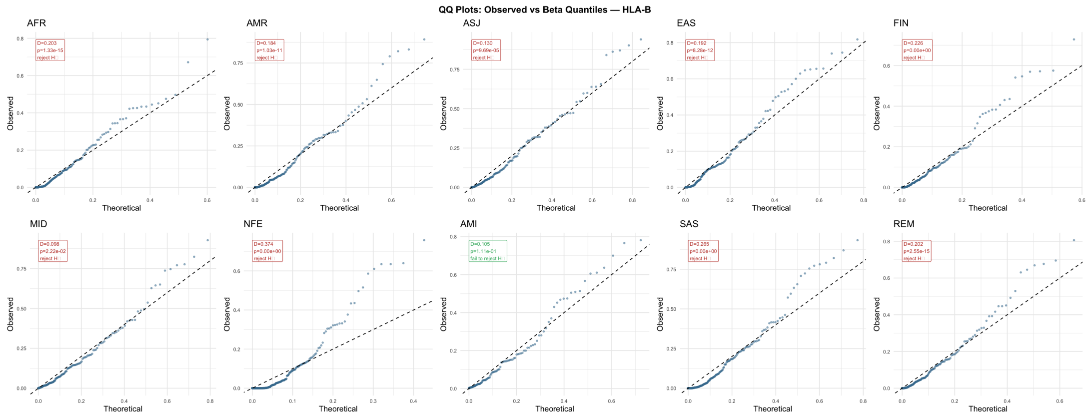


Figure 19: QQ plots for *HLA-B*. The pattern changes qualitatively: points follow the diagonal closely in AMI (consistent with $p = 0.111$), and track it reasonably well in AMR, ASJ, EAS, and MID before diverging at high quantiles. NFE collapses back to the APOE/CYP2C19 pattern due to its large cohort amplifying the ultra-rare spike.

Comparing the three figures reveals the precise nature of the Beta model's failure. For APOE and CYP2C19, the QQ plots show an **L-shaped collapse**: the vast majority of observed quantiles are near zero while the theoretical Beta places them at values between 10^{-3} and 10^{-1}. This means the Beta assigns far too little probability to the lowest frequencies and far too much to intermediate ones. Only a handful of high-frequency variants fall above the diagonal, reflecting the common-variant tail the Beta partially captures.

For HLA-B the picture is more nuanced. In populations with moderate sample sizes (AMI, MID, ASJ), points track the diagonal through much of the distribution before gently diverging at the highest quantiles. This is the expected behaviour of a model that captures the bulk of the spectrum but misses the extremes.

5.3 Summary of Evaluation Findings

Table 9: Summary of goodness-of-fit results per gene.

Gene	Mean D_n	Range D_n	Rejected	Assessment
APOE	0.428	0.374–0.484	9/9	Poor — systematic rightward displacement in all populations
CYP2C19	0.427	0.360–0.465	9/9	Poor — identical structural failure to APOE
HLA-B	0.198	0.098–0.374	9/10	Moderate — acceptable in small founder populations; fails for large cohorts

Three conclusions follow from the evaluation:

1. **The Beta is a poor fit for APOE and CYP2C19.** Mean $D_n \approx 0.43$ and the L-shaped QQ collapse confirm that no two-parameter unimodal distribution can simultaneously describe the sharp ultra-rare spike and the moderate-to-common tail of these genes. This is a structural limitation, not a failure of estimation.

2. **The Beta is a moderate fit for HLA-B.** Lower D_n values (0.098–0.226 for most populations), the single non-rejected fit (AMI, $p = 0.111$), and the near-diagonal QQ patterns for AMI and MID indicate that when balancing selection broadens the frequency spectrum, the Beta becomes an adequate model.

3. **Cohort size confounds the KS test.** NFE consistently produces the highest D_n within each gene not because its AF spectrum is biologically unusual, but because its large sample size ($n > 900$ for APOE, $n > 1500$ for CYP2C19, $n > 1200$ for HLA-B) gives the KS test extreme statistical power to detect even minor deviations from the Beta model.